

VIDHYA SHANMUGAM
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Software Engineer

PROFILE SUMMARY

Software Engineer with 8 years of hands-on experience in full-stack web development, with a strong focus on front-end technologies, UI/UX design, and responsive, user-centric applications. Proven success in delivering scalable software solutions across healthcare and aerospace domains. Played a key role in ISRO satellite system software, including real-time telemetry and launch operations tools. Proficient in PHP, JavaScript, HTML5, CSS3, Bootstrap, and modern development practices. Integrated OpenAI models to automate AR training evaluations, enhancing accuracy and efficiency. Recognized for technical excellence, collaboration, and timely project delivery.

HIGHLIGHTS OF SKILLS

- **Programming Languages:** PHP, JavaScript, TypeScript, C++
- **Frameworks & Libraries:** Angular, Bootstrap, Tailwind CSS, jQuery, Qt
- **Web Technologies:** HTML5, CSS3, AJAX, XML, JSON, Responsive Web Design, SEO, UI/UX Development
- **Servers & Infrastructure:** Windows Server, Linux, IIS, Apache HTTP Server, Nginx, XAMPP, WAMP, LAMP
- **Databases:** MySQL, SQL Server, phpMyAdmin, postgresql, Stored Procedures
- **API & Integration:** REST APIs, Postman, JSON, OpenAPI, OAuth2, JWT, API Testing & Authentication
- **Version Control & DevOps:** GitLab, Bitbucket, SVN
- **Cloud & DevOps:** AWS, Azure, CI/CD, Docker
- **Development Tools & IDEs:** Visual Studio Code, Visual Studio 2015, NetBeans, Sublime Text
- **Project Management & Collaboration:** SharePoint, OpenProject, Agile Methodologies, Sprint Planning, Cross-Functional Team Collaboration
- **Business Intelligence & AI Tools:** Power BI, SSIS, SSAS, Azure OpenAI, OpenAI, Claude, Google Antigravity, Kiro
- **Development Tools:** Visual Studio Code, Eclipse, NetBeans, Sublime Text, Visual Studio 2015
- **Methodologies:** SDLC, Object-Oriented Programming (OOP), Agile, Scrum, Design Patterns, Code Reviews, Unit Testing, Technical Documentation
- **Application Support:** Troubleshooting, Root Cause Analysis, Performance Optimization, Production Support
- **Domain Expertise:** Aerospace Software, Satellite Telemetry Systems, Real-Time Data Dashboards, Healthcare Web Applications
- **Soft Skills:** Problem Solving, Critical Thinking, Communication, Leadership, Time Management, Attention to Detail, Multi-Project Management

PROFESSIONAL EXPERIENCE

Software Engineer

Apr 2021 – Apr 2026

Omega Healthcare Management Services, Bangalore, India

- Developed and maintained 10+ custom web applications using JavaScript, HTML, CSS, and Bootstrap, increasing internal tool adoption by 45% and reducing manual processing time by 30%, showcasing attention to detail and user-focused design.
- Integrated OpenAI language models to automate assessment of AR user audio notes, comparing with reference responses to improve scoring consistency by 40% and enhance training feedback by applying strong problem-solving skills and AI integration knowledge.
- Designed and implemented live dashboards and real-time reporting tools using HTML Canvas and Highcharts, improving visibility into key operational metrics and accelerating decision-making by 30% through effective data presentation.
- Led and mentored a team of three junior developers, using CAST Highlight code reader for static code analysis to enforce best practices, improve code quality, and reduce post-deployment issues by 50%, demonstrating strong leadership, communication, and effective team coordination.
- Delivered technical support and rapid issue resolution across applications, maintaining 98% uptime and increasing system reliability and user satisfaction by 25%, requiring excellent verbal communication and troubleshooting abilities.

Software Developer

July 2017 – May 2020

Indian Space Research Organisation (ISRO), Bangalore, India

- Developed and deployed a responsive front-end UI for the ACSS Tracker used by 50+ satellite engineers. Reduced manual tracking time by 40% (~200 hours/month) by creating a unified feature map and automating status updates, demonstrating UI/UX design and front-end architecture skills.
- Engineered robust backend systems using PHP and MySQL to process and manage over 2 TB of satellite telemetry data, increasing data retrieval speed by 60% and ensuring performance stability during high-pressure launch simulations.
- Played a key role in the Chandrayaan mission by enabling real-time display and interaction with live satellite telemetry, reducing pre-launch validation time by 25%, and demonstrating strong attention to detail and teamwork.
- Integrated version traceability and feature-observation mapping across 10+ ACSS software iterations, increasing debugging efficiency and compliance tracking by 35% applying strong organization skills.
- Collaborated with cross-functional teams to simulate and validate launch workflows, reducing deployment errors by 70%, and significantly enhancing overall system reliability, requiring effective teamwork, problem solving and clear communication.

EDUCATION

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| • <u>Master of Computer Applications (MCA)</u>
Bharathiar University, Coimbatore, India | June 2014 – Apr 2016 |
| • <u>Bachelor of Computer Applications (BCA)</u>
Avinashilingam University, Coimbatore, India | June 2011 – Nov 2014 |

CERTIFICATIONS

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| • IBM
<u>Web Development Fundamentals</u> | 2026 |
| • Vanderbilt University
<u>Prompt Engineering for ChatGPT</u> | 2024 |
| • IBM
<u>Introduction to Structured Query Language (SQL)</u> | 2024 |
| • IBM
<u>SQL: A Practical Introduction for Querying Databases</u> | 2023 |
| • W3Schools
<u>Bootstrap 5 Tutorial</u> | 2023 |
| • University of Michigan
<u>Building Web Applications in PHP</u> | 2022 |